Predicting Mortality in Heart Failure Patients Using Clinical Variables: A Machine Learning Benchmark Study

Abstract

Heart failure is associated with substantial mortality, yet prognosis remains heterogeneous across patients. This study analyzed the Heart Failure Clinical Records dataset for the binary endpoint `DEATH_EVENT` using a combined framework of descriptive analysis, univariate testing, correlation analysis, feature importance ranking, and predictive modeling. Across the multi-angle factor analysis, follow-up time, ejection fraction, and serum creatinine emerged as the most prominent variables, although follow-up time was interpreted cautiously because of its connection to outcome ascertainment. In the primary model benchmark, XGBoost achieved the highest accuracy (0.8333), precision (0.8462), and F1-score (0.6875), whereas random forest achieved the highest AUC (0.8960). In the preprocessing sensitivity analysis, random forest and XGBoost remained relatively stable across raw, winsorized, and scaled data variants. In addition to comparative benchmarking, the selected best-discriminating model was used to generate an individualized mortality probability for a new patient, illustrating the feasibility of patient-level probabilistic inference within the proposed framework. These results suggest that ensemble methods offer robust discrimination in this small tabular dataset, while complementary factor analysis helps identify clinically interpretable prognostic signals.

1 Introduction

Heart failure remains a major cause of hospitalization, recurrent decompensation, and premature mortality worldwide, and it continues to impose a substantial clinical and public health burden [16, 2]. Prognosis in heart failure is highly heterogeneous, reflecting the combined influence of demographic characteristics, cardiac functional impairment, comorbidity, and laboratory abnormalities [3, 13]. For this reason, accurate assessment of mortality risk is important in routine care, where it may inform treatment intensification, follow-up planning, monitoring priorities, and broader clinical decision-making [13, 3]. Beyond estimating overall outcome probability, prognostic analysis is also valuable for clarifying which patient characteristics are most strongly associated with adverse events, thereby supporting a more structured understanding of disease severity and risk stratification [16, 2].

Machine learning has been increasingly adopted in healthcare research for tasks such as diagnosis support, outcome prediction, and patient stratification [1]. In comparison with conventional regression-based approaches, machine learning models can offer greater flexibility for capturing nonlinear relationships, threshold effects, and higher-order interactions among clinical variables [21, 17]. Within heart failure research, these methods have been used to predict mortality, readmission, and other clinically relevant outcomes from routinely collected tabular data [21, 17, 19, 15, 7]. At the same time, predictive modeling alone is often insufficient in medical applications [8, 20]. Clinically useful studies must also address interpretability by identifying the variables that consistently contribute to prognosis, ideally through complementary analytical perspectives rather than reliance on a single model-specific importance measure [18, 5].

Despite the growing literature, several methodological issues remain insufficiently resolved. Many studies report results for only a narrow subset of algorithms, limiting the ability to judge whether observed performance is robust to model choice [22, 11]. In addition, the effect of

preprocessing decisions, including scaling and outlier handling, is often not examined systematically, even though such decisions may affect linear models, neural networks, and tree-based methods in different ways [22, 8]. A related challenge concerns the identification of influential risk factors in small clinical datasets, where limited sample size may lead to unstable estimates, overfitting, and model-dependent conclusions [7, 20]. Under these conditions, a more careful framework is needed that combines comparative prediction with multi-angle assessment of variable relevance using statistical testing, correlation structure, and model-based feature importance [8, 18, 5].

In this study, these issues are addressed through an integrated analysis of heart failure mortality prediction using logistic regression (LR), support vector machine (SVM), k-nearest neighbors (KNN), random forest (RF), extreme gradient boosting (XGBoost), and multilayer perceptron (MLP) models [14, 9, 10, 4, 6, 12]. Predictive performance was benchmarked across models, and preprocessing sensitivity was also examined to evaluate the effect of alternative data representations in a small-sample setting [22, 11]. In parallel, factors associated with `DEATH_EVENT` were examined from multiple analytical perspectives, including descriptive comparison, univariate statistical testing, correlation analysis, and feature importance estimation, in order to identify clinically meaningful variables that showed consistent prognostic relevance. By combining model benchmarking with interpretable risk-factor analysis, this study aims to provide a methodologically transparent and clinically grounded assessment of mortality prediction in heart failure [7, 8].

2 Data Characterization

2.1 Dataset and Variables

The analysis was conducted on the Heart Failure Clinical Records dataset, which contains 299 patient records and 13 variables. The study endpoint was the binary outcome `DEATH_EVENT`, comprising 203 survivors (67.89%) and 96 deaths (32.11%). The available predictors represent routinely collected demographic, comorbidity, laboratory, and follow-up-related information. Continuous variables were age, creatinine phosphokinase, ejection fraction, platelets, serum creatinine, serum sodium, and follow-up time. Binary variables were anaemia, diabetes, high blood pressure, sex, and smoking.

2.2 Preprocessing and Outlier Handling

Initial data inspection showed no missing values. Because several laboratory and physiological measurements spanned wide value ranges, outliers were examined using an interquartile range criterion. Notable extreme values were identified in creatinine phosphokinase, serum creatinine, platelets, serum sodium, and ejection fraction. These observations were retained in the primary analysis because they may represent clinically meaningful severity rather than simple measurement error.

To evaluate whether extreme values materially influenced model behavior, a winsorized version of the dataset was created only for sensitivity analysis. Winsorization was therefore used as a robustness check rather than as the default preprocessing strategy. In addition, continuous variables were standardized for scale-sensitive models. This design separated the main analytical pipeline from robustness comparison and reduced ambiguity in later model interpretation.

2.3 Descriptive Statistical Analysis

Descriptive statistics were summarized separately for continuous and categorical variables to characterize the study cohort. The cohort characteristics are reported in Table 1 and Table 2.

Among 299 patients, 203 (67.89%) survived and 96 (32.11%) died during follow-up. Age was 60.83 (11.89) years and ejection fraction was 38.08% (11.83). Follow-up time was 130.26 (77.61) days, with a median of 115 days (interquartile range [IQR], 73–203).

Several continuous variables showed substantial dispersion. Creatinine phosphokinase had a median of 250.0 (IQR, 116.5–582.0) and ranged from 23 to 7861, indicating marked right skew. Platelet count was 263,358.03 (97,804.24), with a range of 25,100 to 850,000. Serum creatinine was 1.39 (1.03) mg/dL, with a median of 1.1 mg/dL (IQR, 0.9–1.4) and a maximum of 9.4 mg/dL. Serum sodium was 136.63 (4.41) mEq/L, with a median of 137 mEq/L (IQR, 134–140).

For categorical variables, anaemia was present in 129 patients (43.14%), diabetes in 125 (41.81%), and high blood pressure in 105 (35.12%). Men accounted for 194 patients (64.88%), and 96 patients (32.11%) were smokers. Overall, the cohort showed substantial variation in demographic characteristics, comorbidity burden, and laboratory measures, supporting subsequent predictive and risk factor analyses.

Table 1: Descriptive statistics of continuous variables.

Variable	Mean	Std	Min	Q1	Median	Q3	Max
Age	60.83	11.89	40	51	60	70	95
CPK	581.84	970.29	23	116.5	250	582	7861
Ejection fraction	38.08	11.83	14	30	38	45	80
Platelets	263358.03	97804.24	25100	212500	262000	303500	850000
Serum creatinine	1.39	1.03	0.5	0.9	1.1	1.4	9.4
Serum sodium	136.63	4.41	113	134	137	140	148
Time	130.26	77.61	4	73	115	203	285

Table 2: Distribution of categorical variables.

Variable	Category	Count	Percentage (%)
anaemia	0	170	56.86
anaemia	1	129	43.14
diabetes	0	174	58.19
diabetes	1	125	41.81
high_blood_pressure	0	194	64.88
high_blood_pressure	1	105	35.12
sex	0	105	35.12
sex	1	194	64.88
smoking	0	203	67.89
smoking	1	96	32.11
DEATH_EVENT	0	203	67.89
DEATH_EVENT	1	96	32.11

3 Methods

As illustrated in Fig. 1, the proposed framework adopts a dual-branch design that integrates interpretable risk factor detection with predictive modeling in a unified pipeline. Starting from structured clinical data, the framework first applies data inspection and preprocessing procedures, including confirmation of the absence of missing values, standardization where needed, and a separate robustness comparison using winsorization. It then proceeds along two

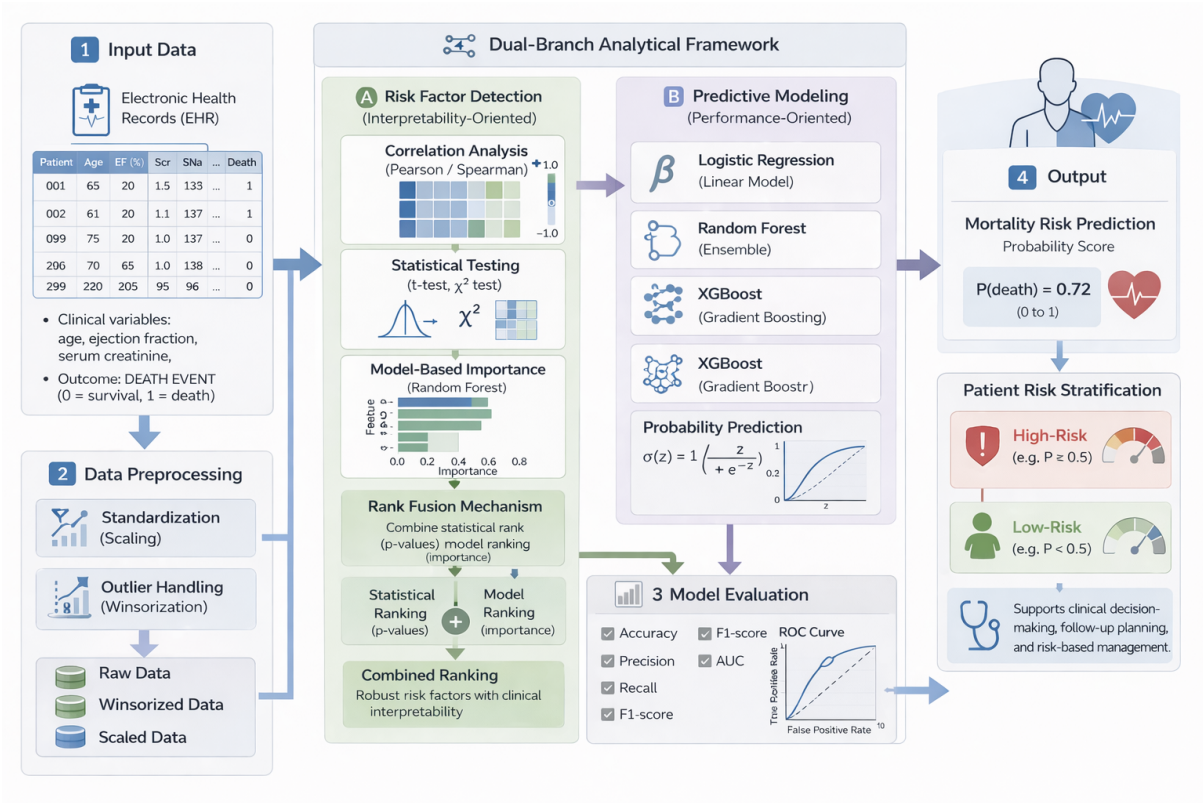

Figure 1: Overview of the proposed dual-branch framework for heart failure mortality prediction, integrating multi-angle risk factor detection with predictive modeling.

complementary branches: one focuses on identifying clinically relevant variables through multi-angle analysis, while the other performs mortality prediction using multiple machine learning models. The outputs of both branches are subsequently evaluated using standard classification metrics to support patient-level risk assessment.

3.1 Risk Factor Detection Method

To identify variables associated with `DEATH_EVENT`, a multi-angle analytic framework was used, combining association analysis, correlation structure, and model-based importance. Pairwise correlations among numeric variables were visualized using a correlation heatmap in order to summarize the overall structure of the dataset and highlight potentially related clinical measurements. Univariate outcome comparison was performed using Welch's t-test for continuous variables and chi-square testing for binary variables, providing a statistical screening step for evaluating whether variable distributions differed between patients who survived and those who died during follow-up.

For a continuous variable, the Welch's t-statistic was defined as

$$t = \frac{\bar{x}_1 - \bar{x}_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}}}, \tag{1}$$

where $\bar{x}_1$ and $\bar{x}_0$ denote the group means, s_1^2 and s_0^2 the group variances, and n_1 and n_0 the group sample sizes for the death and survival groups, respectively. For binary variables, the chi-square statistic was computed as

$$\chi^2 = \sum_i \sum_j \frac{(O_{ij} - E_{ij})^2}{E_{ij}}, \tag{2}$$

where O_{ij} and E_{ij} are the observed and expected cell counts in the contingency table.

Feature importance was estimated using a random forest classifier trained on the full predictor set. Random forest was selected because it can capture nonlinear effects and interactions while providing an interpretable ranking of variable contribution in structured tabular data. Statistical significance and model-based importance were then integrated into a combined ranking procedure. Specifically, each variable received a statistical rank based on ascending p-value from the appropriate univariate test and a model rank based on descending random forest importance. The combined score was defined as:

$$\text{Combined Score}_j = \text{Statistical Rank}_j + \text{RF Rank}_j, \tag{3}$$

where a smaller score indicates greater overall prominence. For variables without a valid univariate test ranking, a penalty rank was assigned to avoid overstating their importance. This strategy allowed the study to identify factors that were supported across complementary analytical perspectives rather than by a single criterion alone. Because the `time` variable reflects follow-up duration and event ascertainment structure, its strong association with mortality was interpreted cautiously and distinguished from baseline biological risk factors.

3.2 Predictive Modeling Method

Mortality prediction was evaluated using six supervised learning models: logistic regression (LR), support vector machine (SVM), k-nearest neighbors (KNN), random forest (RF), extreme gradient boosting (XGBoost), and multilayer perceptron (MLP). Two experiments were conducted. Experiment 1 served as the primary model benchmark and compared all six models on the scaled dataset. This setting was chosen to provide a consistent comparison across model classes while avoiding an unfair disadvantage for scale-sensitive methods. Experiment 2 was a preprocessing sensitivity analysis in which the tree-based models RF and XGBoost were evaluated on raw, winsorized, and scaled versions of the dataset. Because winsorization was introduced only for robustness comparison, the scaled benchmark remained the main reference setting for interpretation.

Logistic regression was included as a conventional and interpretable baseline for binary classification. For patient feature vector $\mathbf{x}$, logistic regression estimates the probability of death as

$$P(y = 1 \mid \mathbf{x}) = \sigma(\beta_0 + \boldsymbol{\beta}^\top \mathbf{x}) = \frac{1}{1 + e^{-(\beta_0 + \boldsymbol{\beta}^\top \mathbf{x})}}. \tag{4}$$

The SVM model was included as a margin-based classifier capable of learning nonlinear decision boundaries through kernel mapping. KNN was included as a distance-based nonparametric baseline in which a patient's class is determined from the labels of nearby samples in feature space. Both methods are sensitive to feature scale, and therefore they were evaluated only in the scaled-data benchmark.

For SVM, the decision function can be written as

$$f(\mathbf{x}) = \text{sign}\left(\sum_{i=1}^{n} \alpha_i y_i K(\mathbf{x}_i, \mathbf{x}) + b\right), \tag{5}$$

where α_i are learned coefficients, y_i are class labels, $K(\mathbf{x}_i, \mathbf{x})$ is the kernel function, and b is the bias term. This formulation allows the classifier to separate classes by maximizing the margin in a transformed feature space.

For KNN, the predicted class for a sample $\mathbf{x}$ is determined by majority vote among its k nearest neighbors:

$$\hat{y}(\mathbf{x}) = \text{mode}\{y_i : \mathbf{x}_i \in \mathcal{N}_k(\mathbf{x})\}, \tag{6}$$

where $\mathcal{N}_k(\mathbf{x})$ denotes the set of the k closest training samples to $\mathbf{x}$ under the chosen distance metric. This method provides a simple reference model based on local similarity structure in the feature space.

Random forest was used as a tree-based ensemble method that aggregates predictions from multiple decision trees and is well suited to tabular clinical data with nonlinear relationships. The final prediction probability can be expressed as the average output across B trees:

$$\hat{P}_{RF}(y = 1 \mid \mathbf{x}) = \frac{1}{B} \sum_{b=1}^{B} T_b(\mathbf{x}), \tag{7}$$

where $T_b(\mathbf{x})$ denotes the probability estimate from the bth tree.

XGBoost was included as a gradient boosting method that builds an additive ensemble of weak learners in a stage-wise manner. Its prediction function can be written as

$$\hat{y}_i = \sum_{k=1}^{K} f_k(\mathbf{x}_i), \qquad f_k \in \mathcal{F}, \tag{8}$$

where each f_k is a regression tree and $\mathcal{F}$ denotes the space of allowable trees. This framework is effective for learning nonlinear decision boundaries while controlling model complexity through regularization.

The MLP model was used as a simple neural-network baseline. For one hidden layer, the mapping can be expressed as

$$\hat{y} = \sigma\left(\mathbf{W}_2\, g(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2\right), \tag{9}$$

where $\mathbf{W}_1$ and $\mathbf{W}_2$ are weight matrices, $\mathbf{b}_1$ and $\mathbf{b}_2$ are bias terms, $g(\cdot)$ is the hidden-layer activation function, and $\sigma(\cdot)$ is the output sigmoid function. The MLP was included to assess whether a modest neural architecture provided any advantage over classical machine learning models in this small-sample setting.

Model performance was evaluated using accuracy, precision, recall, F1-score, and the area under the receiver operating characteristic curve (AUC). The F1-score is defined as

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{10}$$

4 Results

4.1 Experimental Setup and Evaluation Metrics

Two experiments were conducted. Experiment 1 was designed as the primary model benchmark and compared LR, SVM, KNN, RF, XGBoost, and MLP on scaled data [14, 9, 10, 4, 6, 12].. This setting provided a consistent comparison across model classes while ensuring that scale-sensitive methods were not disadvantaged by variable magnitude differences. Experiment 2 was designed as a preprocessing sensitivity analysis. In this experiment, the tree-based models RF and XGBoost were evaluated across raw, winsorized, and scaled versions of the dataset to assess the extent to which predictive performance changed under alternative data preparation strategies. Because winsorization was introduced only for robustness comparison, the scaled benchmark remained the main reference setting for interpretation.

4.2 Risk Factor Detection Findings

The multi-angle risk factor analysis identified several variables with consistent relevance to `DEATH_EVENT`. The correlation heatmap in Figure 2 provides an overview of pairwise associations among numeric variables and illustrates that the dataset contains several non-negligible

inter-variable relationships rather than a set of fully independent predictors. In particular, ejection fraction and serum creatinine showed meaningful relationships with the outcome-related structure of the dataset, supporting their relevance for subsequent interpretation. At the same time, the heatmap suggests that most predictors are not so strongly correlated as to make their contributions completely redundant, which supports the use of complementary statistical and model-based approaches for factor detection.

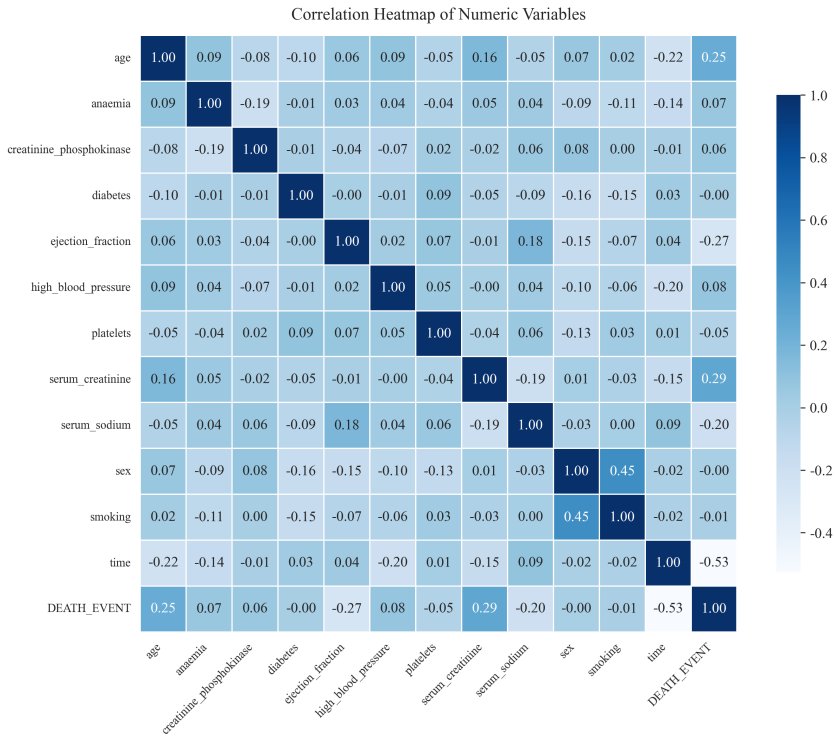

Figure 2: Correlation heatmap of numeric variables used in risk factor analysis.

In the univariate analysis, follow-up time showed the strongest association with outcome, followed by ejection fraction, age, serum creatinine, and serum sodium. No binary variable reached statistical significance in the chi-square analysis. These findings suggest that the measurable prognostic signal in this dataset is concentrated primarily in continuous clinical and laboratory variables rather than in the assessed binary indicators.

The model-based analysis showed a similar pattern. As shown in Figure 3, random-forest-derived feature importance ranked follow-up time, serum creatinine, ejection fraction, age, and creatinine phosphokinase as the most influential variables. The distribution of importance values indicates that follow-up time contributed substantially more than the remaining predictors, while serum creatinine and ejection fraction formed a second tier of clinically interpretable variables with clear prognostic relevance. Age and creatinine phosphokinase also contributed meaningfully, although with smaller importance values.

When statistical ranking and model importance were considered jointly, follow-up time, ejection fraction, and serum creatinine emerged as the most prominent factors overall. Although follow-up time was highly informative for discrimination, its interpretation remains conditional on dataset structure. Greater clinical emphasis should therefore be placed on variables such as ejection fraction and serum creatinine, which are more directly linked to cardiac function and cardiorenal status.

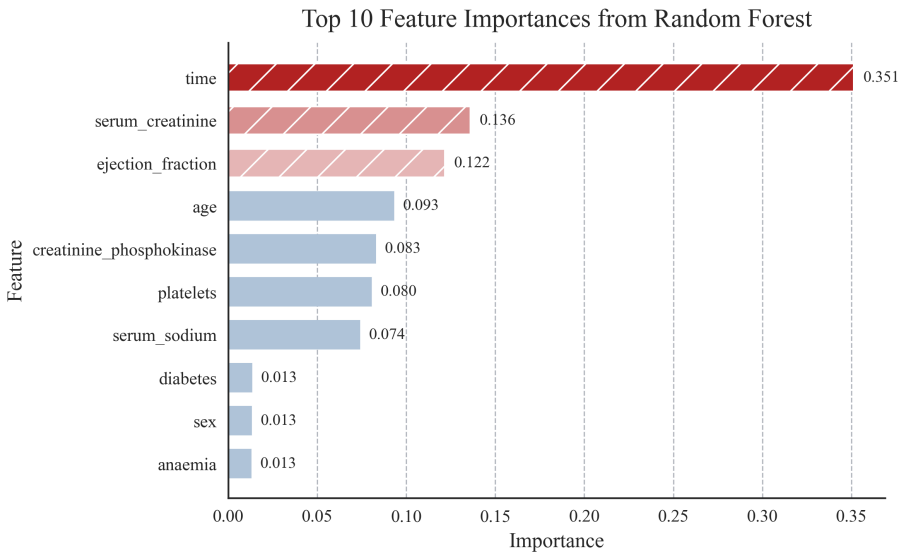

Figure 3: Random-forest-derived feature importance ranking for risk factor analysis.

4.3 Experiment 1: Model Benchmark on Scaled Data

The primary experiment compared six models on scaled data. The strongest overall performance was observed for the tree-based ensemble methods. XGBoost achieved the highest accuracy (0.8333), precision (0.8462), and F1-score (0.6875), whereas random forest achieved the highest AUC (0.8960) with accuracy 0.8167, precision 0.7857, recall 0.5789, and F1-score 0.6667. Logistic regression also reached accuracy 0.8167 and F1-score 0.6667, but with a lower AUC of 0.8614.

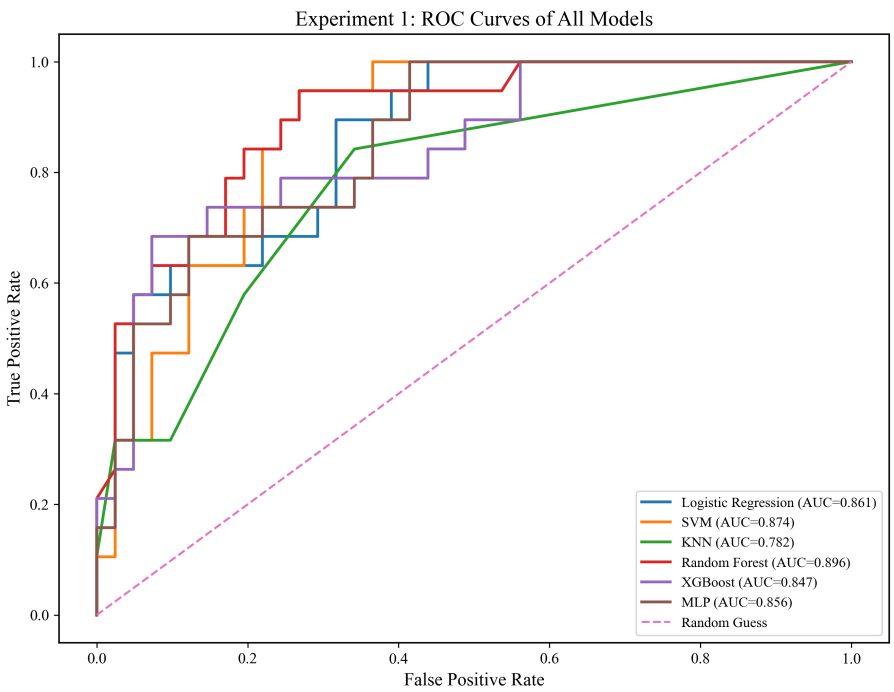

Figure 4: Receiver operating characteristic curves for Experiment 1 model comparison on scaled data.

Among the remaining baselines, SVM achieved accuracy 0.7667, precision 0.6923, recall

0.4737, F1-score 0.5625, and AUC 0.8742. KNN showed the weakest overall results, with accuracy 0.7167, precision 0.6000, recall 0.3158, F1-score 0.4138, and AUC 0.7824. The MLP baseline achieved accuracy 0.8000, precision 0.7333, recall 0.5789, F1-score 0.6471, and AUC 0.8562. Taken together, these results indicate that the two ensemble tree methods produced the most favorable balance between threshold-based classification metrics and discrimination, while the simpler and neural baselines remained competitive but not superior. The ROC curves in Figure 4 illustrate the relative discrimination achieved by the compared methods.

4.4 Experiment 2: Preprocessing Sensitivity Analysis

The secondary experiment examined whether the performance of random forest and XGBoost changed across raw, winsorized, and scaled versions of the data. For random forest, performance varied only modestly across preprocessing settings. Accuracy was 0.8000 on raw data and 0.8167 on both winsorized and scaled data. The corresponding AUC values were 0.8935 for raw data, 0.8864 for winsorized data, and 0.8960 for scaled data. Precision, recall, and F1-score were 0.7692, 0.5263, and 0.6250 on raw data, compared with 0.7857, 0.5789, and 0.6667 on both winsorized and scaled data.

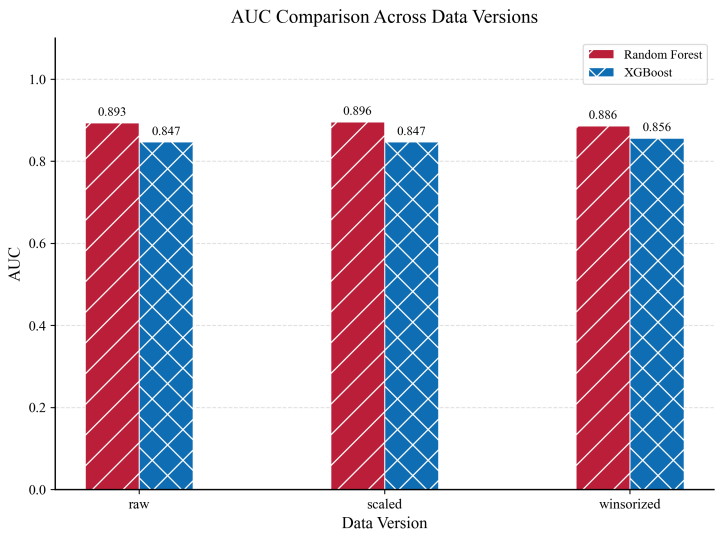

Figure 5: AUC comparison for random forest and XGBoost across raw, winsorized, and scaled data versions.

XGBoost showed similarly stable behavior across preprocessing variants. Accuracy, precision, recall, and F1-score remained unchanged at 0.8333, 0.8462, 0.5789, and 0.6875, respectively, for raw, winsorized, and scaled data. The AUC values were 0.8472 on raw data, 0.8562 on winsorized data, and 0.8472 on scaled data. As shown in Figure 5, the overall ranking of preprocessing strategies was similar across the two models, with no large shifts in discriminative performance.

4.5 Individual Mortality Probability Prediction for a New Patient

To further illustrate patient-level inference, an example prediction was generated for a new patient using the model with the strongest discriminative performance in the benchmark analysis. For this case, the predicted mortality probability was 0.5567, with a corresponding positive class prediction under the default decision threshold. Although presented here as an application example rather than a separate validation experiment, this result demonstrates that the proposed framework can provide individualized probability estimates in addition to cohort-level comparative performance.

Table 3: Example of individual mortality probability prediction for a new patient.

Age	Ejection Fraction	Serum Creatinine	Serum Sodium	Follow-up Time	Predicted Probability	Predicted Class
68	25	1.8	132	90	0.5567	1

In this example, the patient exhibited reduced ejection fraction and elevated serum creatinine, both of which had previously emerged as prominent prognostic variables in the multi-angle factor analysis. The predicted probability is therefore consistent with the broader analytical findings and illustrates how the framework may be used to summarize individualized mortality risk in probabilistic terms.

5 Discussion

This study was designed to address two related questions in heart failure prognosis: which clinical variables appear most strongly associated with mortality, and which predictive models perform most effectively in a small structured dataset. From the modeling perspective, the tree-based ensemble methods showed the strongest overall performance. RF produced the highest AUC, whereas XGBoost achieved the highest accuracy, precision, and F1-score in the main benchmark. Logistic regression remained competitive, while SVM, KNN, and the MLP baseline did not surpass the leading ensemble models. A plausible explanation is that the dataset is small, heterogeneous, and likely to contain nonlinear associations and threshold effects among laboratory and physiological variables. In such settings, ensemble tree methods are often well suited to capturing interactions without strong distributional assumptions. Their relatively stable behavior across preprocessing variants further supports their practical robustness in tabular clinical prediction tasks.

From the risk factor detection perspective, the multi-angle analysis produced a coherent pattern. Variables such as ejection fraction and serum creatinine were supported by both univariate statistical testing and model-based feature importance, which strengthens their interpretation as clinically meaningful markers of prognosis. This finding is medically plausible because ejection fraction reflects cardiac systolic function, whereas serum creatinine reflects renal impairment and the broader cardiorenal burden that is closely linked to adverse outcome in heart failure. The agreement between statistical comparison and model-derived ranking suggests that these variables capture a substantial portion of the mortality-related signal in the dataset.

At the same time, the prominent role of `time` requires careful interpretation. Follow-up time was highly informative in both the statistical and model-based analyses, but it is tied to the duration of observation and the structure of outcome ascertainment rather than functioning as a purely baseline biological determinant of risk. It should therefore be distinguished from clinically actionable baseline predictors when interpreting the combined ranking. Beyond hard class assignment, the framework also provides individualized mortality probabilities for new patients, which may offer a more interpretable summary of model output in practical use. More broadly, the study remains limited by its small sample size, reliance on a single benchmark dataset, and restricted variable set, all of which constrain generalizability and increase the risk of instability in both predictive estimates and factor ranking. The findings should therefore be interpreted as a structured benchmark and exploratory risk-factor assessment rather than as definitive evidence for clinical deployment. Future work should focus on larger multi-center cohorts, external validation, calibration analysis, and clearer separation between baseline predictors and follow-up-related variables.

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